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Review

Functional foods based on extracts or compounds derived from mushrooms

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ABSTRACT

Background: Due to an aging population and illnesses related to current lifestyles, health-related concerns are becoming increasingly more important. Moreover, today's society is more aware of the potential side effects of medicines and is looking for innovative therapeutic alternatives. Hence, the use of natural compounds in the prevention of various diseases and health maintenance has been studied. Among the natural products studied are mushrooms, which are well known for their nutritional value and health-promoting properties. These have been considered as both functional foods and a source of nutraceuticals.

Scope and approach: The present review is aimed at collecting and critically examining current data on the bioactive properties of mushrooms as well as their classification as functional foods and source of nutraceuticals. It also intended to describe the state of the art regarding mushroom formulations currently available on the market, and to highlight what could be done to improve this market in order to make a variety of quality and duly-certified products that promote human well-being available.

Key findings and conclusions: Mushrooms are natural matrices of excellence. Their bioactivity has been proved and therefore, their incorporation in foods has been studied. However, these new food products have not yet gone to market and most of the mushrooms and their compounds are mainly consumed in their natural form or in dietary supplements. Despite interest in such products having grown over the years, in Western countries, mushroom products are not as common as in Asia and legislation needs to be implemented to permit an increase in their consumption.

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1. Introduction

1.1. Functional foods and nutraceuticals

Today's numerous/diverse media channel outlets disclose a wide range of information and generally emphasise the nutrition:health relationship by underlining the need to adopt healthy eating habits. International health organizations recommend 1) limiting energy intake from total fats and 2) shifting fat consumption from saturated to unsaturated fat, while eliminating *trans*-fatty acids. Diets should be rich in fruits, vegetables, legumes, grains and nuts while the intake of free sugars and salt should be limited. By adopting this advice and avoiding sedentary behaviour, it is possible to attain energy balance and a healthy weight as well as to reduce the risk of some diseases (WHO, 2004). The primary role of diet is to provide nutrients essential for metabolic needs but it also prevents malnutrition, promotes satiety and well-being and provides health benefits at physiological levels while ensuring

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optimal health and preventing disease (Alkerwi, 2014).

Given the strong evidence linking diet quality to life quality and due to advances in food science and technology, new food concepts have been developed, including "functional foods" and "nutraceuticals". Although sometimes confused, these terms include related but different concepts. Succinctly, functional foods are conventional or everyday foods consumed as part of the normal daily diet. They should be composed of naturally occurring components (possibly in unnatural concentrations or present in foods which would not normally supply them). Functional foods exert positive effects on target function(s) beyond their nutritive value, enhancing the well-being and quality of life, and/or reducing the risk of disease (El Sohaimy, 2012). Nutraceuticals were first described by DeFelice in 1989 and the definition remains until today. These are defined as "a food (or part of a food) that provides medical or health benefits, including the prevention and treatment of a disease" (El Sohaimy, 2012). Therefore, the role of functional foods is mainly related to reducing the risk of disease rather than preventing it, while nutraceuticals are usually consumed to promote well-being, through the prevention and/or treatment of diseases and/or disorders (El Sohaimy, 2012;; Ghosh, Bagchi, & Konishi, 2014). Nutraceuticals may be a food extract, a single natural compound or nutrient and not necessarily a complete food (*e.g.*, resveratrol, curcumin, vitamin E) which may be included in pharmaceutical form (pills, tablets, etc.) as dietary supplements and as part of a specific diet (Gupta, Chauhan, Mehla, Sood, & Nair, 2010; Sikora, Scapagnini, & Barbagallo, 2010;Silk & Smoliga, 2014).

Sometimes this concept is also misused to describe another food concept – the "food supplement". Food supplements have essentially a feed function, taking the form of medicines (pills or capsules). Once they take the form of medicines and are not consumed as traditional food in the daily diet, they cannot be regarded as functional foods. However, given that from a legal point of view, they are not considered as treating or preventing diseases, they are still governed by food laws (Howlett, 2008). Of note is the fact that functional foods, as conventional foods consumed daily according to a regular food pattern, are not pills, capsules, or have any form of food supplement (EU, 2010). Nutraceuticals, while exerting a pharmacological function, should not be confused with medicines, which are administered in precise doses, under medical supervision, to treat or prevent a specific disease (Howlett, 2008).

Once the functional foods and nutraceuticals have properties inherent to foods and medicines, it is considered that they are located on the border between conventional foods and medicines (Smith & Charter, 2011).

For a better understanding of the concepts of functional foods and nutraceuticals, as well as the related (and sometimes misused) food supplements and medicines, Table 1 shows the main differences between them.

2. Mushrooms "as" and "in" functional foods

2.1. Bioactive properties of mushrooms that confer them the "status" of functional foods

When we think of a balanced diet and the healthiest means by which to achieve it, plants and plant products immediately come to mind as well as mushrooms. Although their use has been reported for thousands of years, it has only been in recent years that the consumption of mushrooms has increased, mainly due to the increasing awareness that a stable and balanced diet exerts a key role in normal body functioning and sustaining health. Consumed for their texture, aroma and flavour, mushrooms are an excellent choice for both their nutritional value and medicinal properties (Valverde, Hernández-Pérez, & Paredes-López, 2015). Indeed, some authors consider mushrooms as "inherent functional foods" (Smith & Charter, 2011). Their chemical composition includes several bioactive compounds which gives them the ability to exert beneficial effects at different levels. These bioactive properties include immunomodulating, antitumour, anti-hypercholesterolemia, antibacterial and antifungal, anti-inflammatory, antiviral, anti-diabetic, and cardiovascular beneficial effects (Valverde et al., 2015; Wasser, 2014). In the supplementary material, Table S1 lists the wide-range bioactive properties reported for different mushroom species.

2.1.1. Antioxidant properties of mushrooms

In the human body (as in other aerobic organisms) free radicals and other reactive species are continually being produced during normal cellular metabolism. This may occur by "accident" (*e.g.*, the leakage of electrons from the mitochondrial electron transport chain) or with some required purposes, such as involvement in cell signalling and homeostasis processes or defence against pathogens (Ferreira, Barros, & Abreu, 2009; Halliwell & Gutteridge, 2015).

As shown in Fig. 1, free radicals have different causes and different cellular targets, namely proteins, carbohydrates, lipids and nucleic acids. For this reason, many disease conditions have been associated with oxidative stress, such as atherosclerosis, diabetes and cancer (Halliwell & Gutteridge, 2015). As is common sense, the antioxidant defences can be endogenous or acquired through the diet. Natural compounds have emerged as key candidates for obtaining antioxidant compounds, due to recognition of the inverse relationship between dietary intake of natural antioxidants and the incidence of human diseases. This has led to an increased worldwide interest in using natural antioxidants in food, cosmetics and the pharmaceutical industries (Taofiq, González-Paramás, Martins, Barreiro, & Ferreira, 2016; Valverde et al., 2015).

Mushrooms have been reported as a source of several antioxidants. These include phenolic compounds, vitamin C and E and carotenoids (Ferreira et al., 2009). Actually, many mushroom extracts have been studied for their antioxidant potential. These studies reported a direct relationship between the compounds

Table 1
General differences between functional foods, nutraceuticals, food supplements and medicines.

	Functional food	Nutraceutical	Food supplement	Medicines
Form	Food	Pill, tablet, capsule, syrup	Pill, tablet, capsule, syrup	Pill, tablet, capsule, syrup, injectable, etc.
Consumption	Consumed as part of the normal diet	Daily consumed, (usually for a period of time) once they are food constituents/extracts	Usually during a certain period of time	Controlled doses according to medical prescription and for a predetermined period of time
Purpose	Exert a health or physiological effect; Improvement of the state of health and well-being and/or reduction of the risk of disease	Promote well-being through the prevention and/or treatment of diseases and/or disorders	Typically consumed to ensure the intake of certain ingredient(s) (*e.g.*, vitamins, minerals, amino acids); They may also help to reduce the risk of disease	Pharmacological purpose; To treat a specific disease

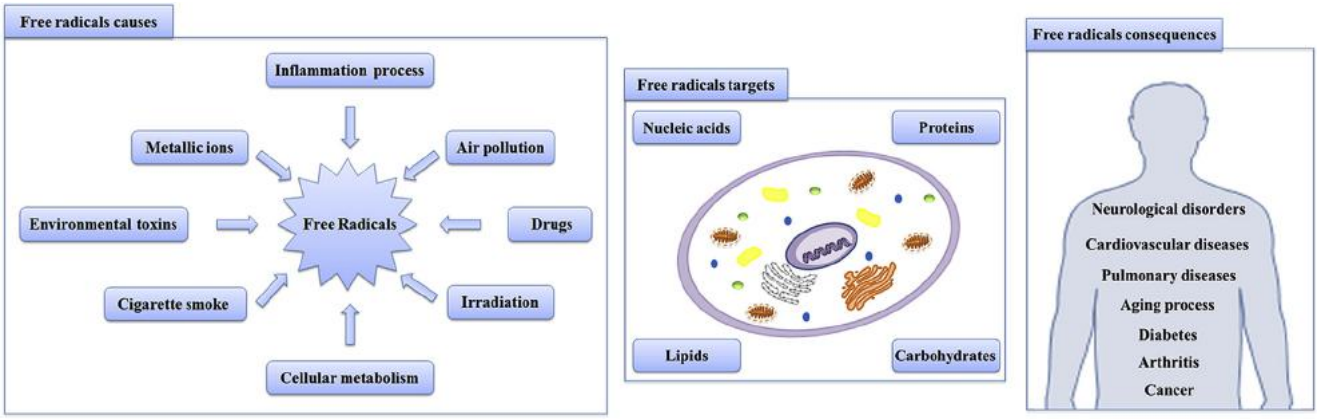

Fig. 1. Main causes for the overproduction of free radicals, probable cellular targets and main consequences in human health associated with oxidative stress (illustration based on the information contained in Ferreira et al., 2009 and Halliwell & Gutteridge, 2015).

present in the extracts (mainly phenolic compounds) and the verified bioactivity. Overall, these studies suggest that mushroom extracts/compounds could be potentially bioactive ingredients for use in the formulation of functional foods as well as natural antioxidants in food systems (Dubost, Ou, & Beelman, 2007; Kaewnarin, Suwannarach, Kumla, & Lumyong, 2016; Kimatu et al., 2017; Tel, Ozturk, Duru, & Turkoglu, 2015).

Although most of the studies about the antioxidant potential of mushrooms used *in vitro* assays, there are some *in vivo* studies. For example, Liu, Jia, Kan, and Jin (2013) reported a study in which an ethanol extract obtained from the species *Agaricus bisporus* (J.E.Lange) Imbach, and rich in gallic acid, protocatechuic acid, catechin, caffeic acid, ferulic acid and myricetin, was administered in mice for 30 consecutive days (300, 600 and 1200 mg/kg body weight per day). As a result of this treatment, the activities of antioxidant enzymes in mice' serums, livers and hearts were significantly enhanced (mainly in animals fed with 600 and 1200 mg/kg body weight per day). Other studies reported also the beneficial interaction between extracts from *Pleurotus* species administered in rats (200 mg/kg body weight per day, for 4 days), and the enzymes involved in the antioxidant defence system (Jayakumar, Ramesh, & Geraldine, 2006; Jayakumar, Sakthivel, Thomas, & Geraldine, 2008; Jayakumar, Thomas, & Geraldine, 2010). Some of these works attributes the antioxidant capacity to lectins present in the extracts (Rana et al., 2012), as well as chrysin, rutin, β-carotene, the antioxidant vitamins ascorbic acid and α-tocopherol, and total phenolic contents (Jayakumar, Thomas, & Geraldine, 2009). Some *Pleurotus* extracts, administered in rats (200 mg/kg body weight, intraperitoneally, for 21 and 30 days) have been also reported as inhibitors of lipid peroxidation in liver, kidney, heart and brain (Jayakumar, Thomas, & Geraldine, 2007) and protein carbonyls (Jayakumar, Thomas, Ramesh, & Geraldine, 2010) in liver and kidney of aged rats. In another study, oxidative stress by carbon tetrachloride (CCl4) was induced in rats. These had been treated with an extract obtained from *Pleurotus ostreatus* (Jacq. ex Fr.) P. Kumm. (200 mg/kg body weight per day, for 4 days), and it was observed a decrease in the concentration of MDA in liver, kidney, heart and brain (Jayakumar et al., 2006; Jayakumar et al., 2008). Nada, Omara, Abdel-Salam, and Zahran (2010) have also reported that the insoluble polysaccharides present in mushrooms prevent CCl-4 hepatotoxicity in rats treated with *P. ostreatus* insoluble non-starch polysaccharides (100 and 200 mg/kg, for 15 days).

All these studies have demonstrated the antioxidant potential of mushrooms, hence the claim of their possible use in bioactive formulations.

2.1.2. Antimicrobial properties of mushrooms

The term antimicrobial comprises a wide variety of pharmaceutical agents which include antibacterial, antifungal, antiviral and anti-parasitic medicines (Leekha, Terrell, & Edson, 2011). The treatment of bacterial infections is extremely important. Despite the wide variety of existing antibiotics (*e.g.*, aminoglycosides, β-lactams, oxazolidinones, lincosamides, quinolones; Gualerzi, Brandi, Fabbretti, & Pon, 2013), treating bacterial infections has become an increasingly complicated task due to the development of bacterial resistance in recent years due to the inappropriate use of antibiotic drugs. Because of this, the development of antibiotics has been one of the most important scientific challenges. Antibiotics exert their effect mainly via 5 ways: 1) By inhibiting the cell wall synthesis interfering with peptidoglycan (cell wall component) synthesis, which will affect the rigidity and integrity of the cell wall leading to cell lysis/death; 2) by altering the function or integrity of the cell membrane, disorganizing its permeability so that nucleic acids and cations leak out and the cell dies; 3) by inhibiting nucleotide synthesis or their interconversion; 4) by preventing DNA from functioning as an appropriate template; and 5) by interfering with the polymerases involved in DNA replication and transcription (Gallo, Lancini, & Parenti, 2013, Fig. 2). Antibiotics also inhibit protein synthesis by interacting directly with the ribosome, binding to one or both 50S and 30S subunits. The metabolism of the cell may also be affected by antimicrobials, such as sulfonamides which interfere with folate metabolism. This class of antimicrobials, competitively block the biosynthesis of tetrahydrofolate, which is necessary for the ultimate synthesis of DNA, RNA and bacterial cell wall proteins.

Pathogenic fungi also cause injury in humans, animals, crops and other living organisms (Alves et al., 2013). Although fungal infections contribute substantially to human morbidity and mortality, and despite the need for efficient diagnostic tests and new medicines and vaccines, research on the pathophysiology of human fungal infections that cause disease falls short of others caused by other pathogens (Brown et al., 2012). Fungal contamination is also closely associated with agricultural products. Among the best known and widely studied moulds that mostly contaminate agricultural products and produce mycotoxins are the *Penicillium*, *Aspergillus* and *Fusarium* genera (Commission Regulation (EC) No 401/2006). Besides their metabolic versatility with great biotechnological potential, having many uses for the production of several food and non-food by-products (*e.g.*, production of the antibiotic

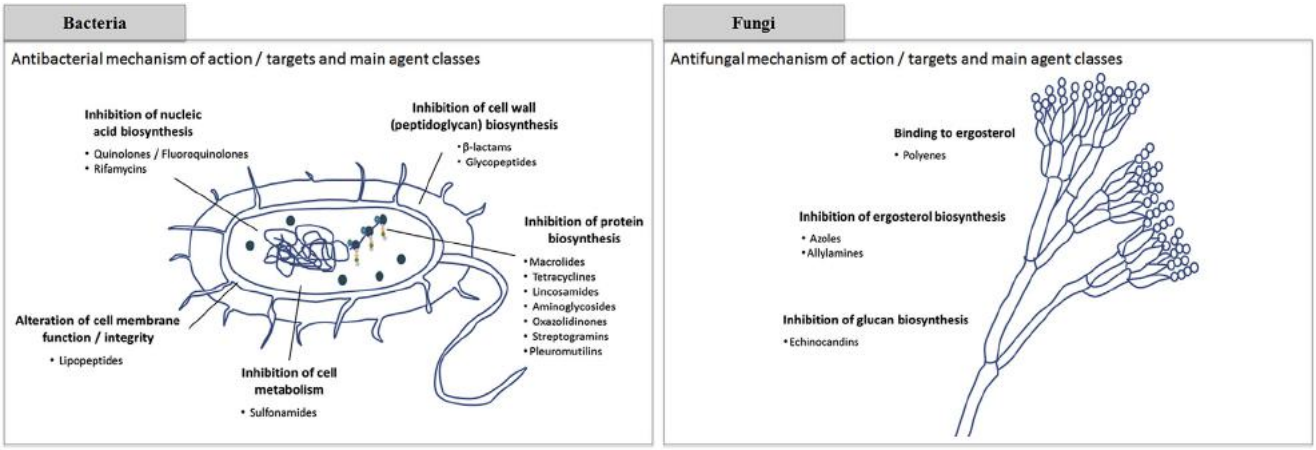

Fig. 2. Mechanism of action of the antibacterial and antifungal agents in clinical use (illustration based on the information contained in Gallo et al., 2013).

penicillin by *Penicillium* species, large-scale obtaining of citric acid carried out by *Aspergillus niger* or *Fusarium venenatum* production to use as a meat substitute), these genera also include the foremost strains producers of mycotoxins (*e.g.*, *Aspergillus flavus* and *Aspergillus parasiticus* producers of aflatoxins; or *Aspergillus ochraceus*, *Aspergillus carbonarius*, and *Penicillium verrucosum* producers of ochratoxin A). It should be noted that guidelines exist to monitor/control the levels of mycotoxins in foodstuffs (Commission Regulation (EC) No 401/2006), to ensure the safety of consumption.

Some studies suggest mushrooms as inhibitors of mycotoxin production. Culture filtrates from *Lentinula edodes* (Berk.) Pegler (also known as *Lentinus edodes*) and T*rametes versicolor* (L.) Lloyd have revealed promising aflatoxin-inhibiting effects (Reverberi et al., 2005; Zjalic et al., 2006).

Antifungal agents, they can form a complex with ergosterol compromising cell membrane fluidity. Other antifungals such as azoles, inhibit the synthesis of this sterol. They are 14α-demethylase inhibitors, an enzyme which catalyzes a critical reaction of the sterol biosynthetic pathway, leading to the depletion of ergosterol in the membrane. Finally, other antifungal classes inhibit glucan biosynthesis, acting as specific non-competitive inhibitors of β-(1,3)-glucan synthetase (Gallo et al., 2013, Fig. 2).

Mushrooms, like other organisms, secrete antimicrobial compounds in order to survive and thrive in the environment. Although this property is proven, antibiotic production is still mainly done using only microfungi (Korzybski, Kowszyk-Gindifer, & Kurylowicz, 2013).

Regarding the antimicrobial activity of mushroom fruiting bodies, some work has already reported the compounds involved in conferring this property. These include: 1) low-molecular weight (LMW) compounds such as terpenes, steroids, anthraquinones, benzoic acid derivatives and quinolones; and 2) high-molecular weight (HMW) compounds, mainly peptides and proteins (Alves et al., 2012).

The antimicrobial activity of mushrooms is assessed through different methodologies. However, these methodologies evaluate the capacity of the mushroom compounds to inhibit pathogens growth.

Mushrooms have been reported to have greater antibacterial activity against Gram-positive bacteria. The most widely cultivated and consumed mushroom worldwide is *Agaricus bisporus* (J.E. Lange) Imbach (champignon), a species with considerably high bioactivity against various species. The potential of this mushroom so widely consumed throughout the world is probably not known to all consumers/enthusiasts. However, it proves that the bioactive potential of these natural products is comprehensive, from the most widely known species to species less able or unable to be produced (Alves et al., 2012).

Gram-positive bacteria strains against which mushrooms are effective include *Bacillus subtilis*, *Bacillus cereus*, *Listeria monocytogenes*, *Micrococcus luteus*, *Micrococcus flavus*, MRSA, *Staphylococcus* sp., and *Staphylococcus epidermidis* (Alves et al., 2012).

Among all studied mushrooms species, *L. edodes* (known as shiitake) has received the greatest attention and seems to exert a broad antimicrobial action against both Gram-positive and Gram-negative bacteria. The latter include *Escherichia coli*, *Klebsiella aerogenes* and *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, or *Salmonella* sp (Alves et al., 2012).

Although a considerable number of studies illustrate the antimicrobial activity of mushrooms, information relating to its mechanism of action is scarce. Thus, detailed studies on the mechanism of action should also be made with these matrices.

The abovementioned LMW and HMW compounds from mushrooms are also reported as having antifungal properties. The fungal pathogens against which the mushrooms are effective include the genus *Candida*, comprising the species *Candida albicans*. However, mushrooms can be also effective against many other pathogenic species such as *Aspergillus* sp. (*i.e. A. fumigatus*, *A. niger*, *A. versicolor*, *A. ochraceus*), *Penicillium* sp. (*i.e. P. funiculosum, P. ochrochloron, P. verrucosum*) or *Trichoderma víride* (Alves et al., 2013). Among the most widely studied and bioactive species against fungi are the mushroom *A. bisporus*, the *Oudemansiella canarii* (Jungh.) Höhn. (reported as a pantropical species), the *Schizophyllum commune* Fries (a species mainly consumed in Mexico and in the tropics) and the widely known *Ganoderma lucidum* (Curtis) P. Karst (Alves et al., 2013).

However, and as for the antimicrobial activity assays, most of the studies available assessing the antifungal activity of mushrooms do not include clarification of the mechanism of action.

2.1.3. Immunomodulating and antitumour effects of mushrooms

The immune system consists of a network of cells, tissues and organs that work together to defend the body against attacks by "foreign" invaders (National Institute of Allergy and Infectious Diseases, 2003 and National Cancer Institute, 2015).

Consumers seek to promote the function of the immune system and thus help strengthen the body's own natural defences through a healthy diet. It is well known that the immune system can be modulated by foods, supplements or endogenous bioactive agents. This modulation may be achieved by substances that act as

immunosuppressants, immunostimulants or, as is usually with natural products, in order to balance the immune function (stimulating some functions and suppressing others) (Point Institute, 2013).

Mushrooms influence immunity through the activation of components from the innate immune system such as natural killer cells (NK Cells), lymphocytes, neutrophils or macrophages, inducing the secretion of pro-inflammatory cytokines such as TNF-α (tumour necrosis factor-alpha), IFN-γ (interferon – gamma), IL-1β (interleukin –1 beta), IL-10 (interleukin 10) or IL-12 (interleukin 12). These cytokines activate the adaptive immunity through the promotion of B cells for antibody production and stimulation of T cell differentiation to T helper (Th) 1 and Th2 cells, which mediate cell and humoral immunities (El Enshasy & Hatti-Kaul, 2013). Moreover, mushrooms and/or their bioactive compounds may also exert this effect by promoting increased maturation and function of the most important antigen-presenting cells, DCs (dendritic cells) (Zhong et al., 2017). DCs have a strong stimulatory activity on T cells. Consequently, stimulating the production and maturation of DCs promotes the regulation of T cell dependent immunity (Ricciardi-Castagnoli, 2013).

Fig. 3 represents the key players in an immune response against the main pathogens to which our body is exposed. Bone marrow, thymus, spleen and lymph nodes are essential for the normal functioning of the human immune system being connected by a network of lymphatic vessels in which lymphatic fluid moves from organ to organ. This lymphatic fluid is rich in white blood cells/leukocytes which play an important role in the human body's response to infection. Tonsils and adenoids are also part of the lymphatic system helping to protect against pathogens entering through the mouth or nose. Provided the physical barriers (*i.e.* skin and epithelial tissues) are broken, the next line of the body's defence which involves white blood cells, is activated. The cells involved in this response are granulocytes, *i.e.* basophils, eosinophils and neutrophils. Monocytes have a short half-life in circulating blood before they migrate and attach to various tissues and differentiate into macrophages (*e.g.*, Kupffer's cells in the liver or alveolar macrophages in the lungs), the major type of white blood cells involved in innate immunity in tissues. They also function as antigen-presenting cells (APCs) for helper T lymphocytes. Mast cells are also granulocyte cells, well known for their involvement in allergic and anaphylactic reactions and can be found in many tissues such as skin, connective tissues, the spleen or lymph nodes. These cells also synthetize and secrete lipid mediators such as prostaglandins as well as cytokines which act as messengers in regulating immune response and stimulating adaptive immunity. Dendritic cells are very versatile cells and immature forms can digest pathogens by phagocytosis. However, since they are also APCs, they have developed the means to preserve useful information from the ingested particles which serves to initiate adaptive immunity. They migrate to the lymph nodes, where the matured cells present the processed antigen (on their surface) to the T cells. Therefore, dendritic cells are important in immune response by bridging the innate and acquired immunity through shifting from antigen-capturing cells to antigen-presenting cells. Natural Killer (NK) cells are a type of white blood cell that can recognize infected and abnormal cells, such as cancer cells and kill these by inducing them to undergo apoptosis or by producing cytokines, such as interferon-gamma (IFN-γ). This in turn activates macrophages to become more effective at killing phagocytosed microbes. Lymphoid cells, namely T- and B-lymphocytes are the white blood cells responsible for acquired immunity. B cells have the capacity to produce antibodies, distinguishing the body's own cells (self) from foreign cells (nonself) and eliminating infected and tumour cells (this last part carried out by killer T cells) (Abbas, Lichtman, & Pillai, 2012).

The immunomodulatory properties of mushrooms may also be related to their antitumour potential and some compounds which exert immunomodulatory effects, also show antitumour activity. Examples of mushroom species having immunomodulatory properties are: *Agaricus blazei* Murrill, *Cordyceps sinensis* (Berk.) Sacc., *G. lucidum*, *Grifola frondosa* (Dicks.) Gray and *Hericium erinaceus* (Bull.) Persoon. The main classes of compounds with immunomodulatory properties are terpenes and terpenoids, lectins, fungal imunommodulatory proteins (FIPs) and polysaccharides (particularly β-D-glucans, as well as polysaccharopeptides and polysaccharide proteins) (El Enshasy & Hatti-Kaul, 2013).

From the most studied bioactive compounds derived from mushrooms with immunomodulatory and antitumour activity, we highlight polysaccharides. Some of the best known polysaccharides are lentinan, isolated from shiitake (*L. edodes*) and schizophyllan, isolated from *Schizophyllum commune* Fries. Due to their

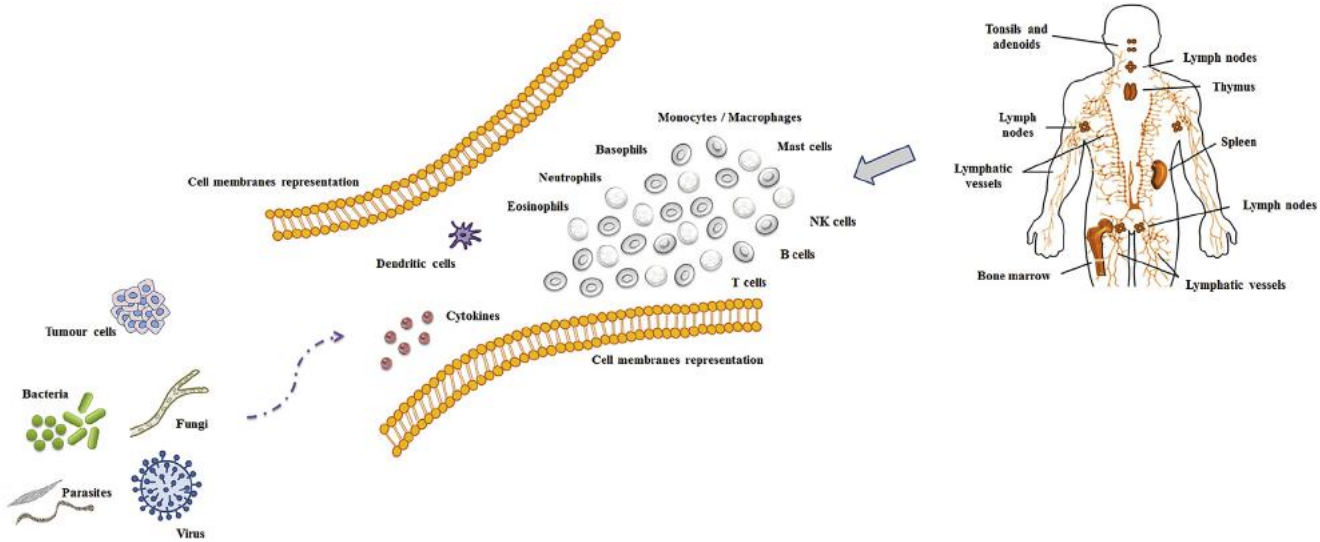

Fig. 3. Overview of the key intervenient in the immune response, with the main antigens to which our body is exposed and major organs and cells involved in the response (illustration based on the information contained in Abbas et al., 2012).

immunomodulatory properties, these β-1,3-D-glucans with β-1,6 branches have been licensed and approved in Japan since 1986 for gastric cancer (lentinan) and head and neck cancer (schizophyllan) treatment. Moreover, they may be used in the clinic in combination with chemotherapy (Ina, Kataoka, & Ando, 2013; Ngwuluka, Ochekpe, & Aruoma, 2016).

From the polysaccharide conjugates showing immunomodulatory properties the following stand out: 1) polysaccharopeptide krestin (polysaccharide-K, PSK) and 2) polysaccharopeptide PSP -, both isolated from *T. versicolor*, from the "CM-101" strain - and the latter from the "Cov-1" strain. These polysaccharide conjugates are chemically similar, heteroglucans with α(1–4)- and β(1–3) glycosidic linkages with a protein component, with the presence of fucose in PSK and rhamnose and arabinose in PSP distinguishing the compounds (Wu, 2015). Several studies have been performed with PSK. This compound is commercially produced in Japan and was approved for use in 1977 (Cui & Chisti, 2003). PSK has been reported to inhibit the development of cancer metastasis through the prevention of adhesion, invasion, motility and metastatic growth of tumour cells, as well as being an inducer of apoptosis in lymphoma, leukemia and pancreatic cells (Hattori, Komatsu, Shichijo, & Itoh, 2004). Some clinical studies have evaluated the potential use of PSK as an adjuvant to conventional cancer therapies. This compound in conjunction with chemotherapy was shown to increase survival rates of patients after curative gastric cancer resection (Oba et al., 2007) and patients with colon cancer (Yamashita et al., 2007). Another study found that PSK improves gene expression of several inflammatory cytokines such as TNF-α, IL-1, IL-8 and IL-6 *in vivo* or *in vitro*. These cytokines, produced by monocytes, macrophages and other cell types, mediate multiple biological effects by directly stimulating the cytotoxic T-cells against tumours and enhance antibody production by B lymphocytes and IL-2 receptor expression on T-lymphocyte (Kato et al., 1995). The induction of TNF-α and IL-6 may also be due to the fact that constituents from PSK act as ligands for toll-like receptors-4 (Price, Wenner, Sloper, Slaton, & Novack, 2010).

PSP, a Chinese product commercially available since 1987 (Cui & Chisti, 2003), has been documented to improve the quality of life in cancer patients by providing substantial pain relief and enhancing immune status in 70–97% of patients with stomach, esophagus, lung, ovary and cervical cancers. Both PSK and PSP boosted immune cell production, ameliorated chemotherapy symptoms and enhanced tumour infiltration by dendritic and cytotoxic T-cells (Kidd, 2000).

Among the polysaccharide-protein complexes, PSPC from *Tricholoma lobayense* Heim is one the most well-known glycoproteins. This complex can restore and improve the phagocytic function of macrophages in tumour-bearing mice (Moradali, Mostafavi, Ghods, & Hedjaroude, 2007). A proteoglycan which is also well-known for its immunomodulatory and antineoplastic properties is AbM, from the mushroom *A. blazei*. AbM contains among others, large amounts of polysaccharide compounds such as β-1,6-glucan, α-1,6- and α-1,4-glucan, glucomannan and β-1,3-glucan (Biedron, Tangen, Maresz, & Hetland, 2012).

As previously mentioned, some proteins present in mushrooms also have immunomodulatory properties. These include lectins and immunomodulatory proteins (FIPs) (Xu, Yan, Chen, & Zhang, 2011; El Enshasy & Hatti-Kaul, 2013). In fact, lectins from *Volvariella volacea* (Bul. ex Fr.) Singer exhibit stronger immunomodulatory activity through the activation of lymphocytes by successive calcium influx, nuclear localization of NFAT transcription factor, induction of activation markers, CD25 and CD69, intracellular cytokine production and cell proliferation (Sze, Ho, & Liu, 2004). Other lectins (TML-1 and TML-2) from *Tricholoma mongolicum* S. Imai have also been isolated and shown immunomodulatory properties through the stimulation of nitrite production and tumour necrosis factor (TNF)-α, and inhibition of the growth of mouse lymphoblast-like (p815) mastocytoma cells by the production of macrophage-activating factors (*i.e.* interferon (IFN)-γ and other cytokines, activated through upregulation of inducible nitric oxide synthase (NOS), interleukin (IL)-1β and transforming growth factor-β) (Liu, Ng, Wang, Fung, & Ooi, 2005). Other lectins have also been isolated from other mushroom species such as *G. frondosa*, *Ganoderma capense* (Lloyd) Teng and *Clitocybe nebularis* (Batsch), P.Kumm. Regarding FIPs, the best known is Ling-Zhi-8 from *G. lucidum* which acts as an immunosuppressive agent (El Enshasy & Hatti-Kaul, 2013).

Numerous studies have confirmed the antitumour potential of extracts and compounds obtained from different mushroom species (Ferreira, Vaz, Vasconcelos, & Martins, 2010; Giavasis, 2014). Moreover, many other studies have been undertaken to identify mushroom extracts/fractions/compounds with antitumour potential and further develop nutraceutical and/or pharmaceutical formulations. Regarding programmed cell death (PCD) induction, several species have shown potential, particularly as modulators of apoptosis (Reis et al., 2016; Vaz, Almeida, Ferreira, Martins, & Vasconcelos, 2012a; Vaz et al., 2012b) and autophagy (Oliveira et al., 2014; Reis, Lima, Morales, Ferreira, & Vasconcelos, 2015).

We can therefore conclude that the bioactivity of mushrooms and their compounds are not watertight. For example, the antitumour activity can be based on the activation of the immune system. Also, the attempt to eliminate free radicals may be based on immune response activation. Hence, antitumour or antioxidant potential associated with immunomodulating properties can exist with some compound(s).

2.1.4. Other bioactive properties of mushrooms

While most studies on the bioactivity of mushrooms are focused on the above described properties, other beneficial health properties have also been assigned to these natural products. The antiviral activity of mushrooms has also been highlighted over the years. Some studies describe extracts or even isolated compounds from mushrooms with antiviral activity. The mechanism of antiviral action may be direct (e.g. inhibiting the viral enzymes) or indirect as a result of the immunostimulating activity of polysaccharides or other complex molecules (Brandt & Piraino, 2000).

Besides, other properties such as anti-inflammatory, hypoglycemic and hypocholesterolemic which result in the prevention of cardiovascular diseases have also been attributed to mushrooms (Valverde et al., 2015). As supplementary material, Table S1 compiles data from the majority of studies with edible and/or medicinal mushrooms reported as having such properties. This Table includes those species not widely consumed (due to poor palatability) but excludes toxic species.

Given the information set out above, the position of certain authors who consider mushrooms as "inherent functional foods" is understandable.

Today, mushrooms are consumed in various forms, as foods, in the form of dietary supplement DS), or as a medicine, usually called "mushroom pharmaceuticals". Moreover, they are used as natural bio-control agents in plant protection (acting as insecticides, fungicides, bactericides, etc.) and in cosmetics (due to their film-forming capability, antioxidant, anti-allergic or antibacterial activities, stimulation of collagen activity, among others) (Taofiq et al., 2016; Wasser, 2014).

2.2. The use of mushroom extracts and compounds in functional foods

Although much information exists on the pharmacological/

medical properties of mushrooms and their components (section 2.1) their incorporation in foods such as components for functional foods has not been comprehensively studied even though this is an important field of knowledge. From a commercial point of view, this entails the possibility of obtaining new foods with new flavours and which are functionalized. From the standpoint of food itself, it is indispensable to know all possible interactions that could promote or inhibit the bioactivity allocated to each extract/compound. It is well known that the bioactivity associated with an extract/compound may be affected by the processing activities or by other components present in food (Giavasis, 2014). For example, it is described that ganoderan, a typical β-(1,3) glucan isolated from the water-soluble fraction of the fruit body or from the filtrates of liquid cultures of *G. lucidum* mycelia, is degradable by pectinases and dextranases (Giavasis, 2013, pp. 413–468). Therefore, its bioactivity may be compromised in foods with such enzymes. Carrageenan, a polymer used in food industry as a gelling agent, thickener and stabilizer, was also reported to interfere with the antitumour activity of some polysaccharides, such as lentinan (Giavasis, 2013, pp. 413–468). Although information remains very scarce, which recognizes some obstacles in the inclusion of certain compounds extracted from mushrooms in foods (to enhance their flavour, preserve or functionalize them), some promising results do exist.

Today, it is very common to find on the market several products supplemented with dietary fibre, to give rise to fitness-promoting foods (low in calories, cholesterol and fat). Wheat, oats, barley and rice are often used to enrich baked and extruded cereal products with fibre. Since mushrooms are natural products rich in fibre, they are also being considered an excellent choice to replace the most commonly used cereal flours. Ulziijargal, Yang, Lin, Chen, and Mau (2013) used mushroom mycelia of *Antrodia camphorata* (M.Zang & C.H.Su) Sheng H.Wu, Ryvarden & T.T.Chang, *A. blazei*, *H. erinaceus* and *Phellinus linteus* (Berkeley & Curtis) Teng to substitute 5% of wheat flour to make bread. With this work they concluded that the incorporation did not adversely affect bread's texture and mycelium supplemented bread showed higher umami intensity than white bread. Moreover, the incorporation may have beneficial health effects since the supplemented bread contains substantial amounts of the amino acids γ-aminobutyric acid (GABA) and ergothioneine.

As previously reported, polysaccharides are one of the most widely studied and recognized potential bioactive molecules derived from mushrooms. Although their potential has been recognized over the years, these compounds used in the food industry are mainly produced by microorganisms – bacteria and microfungi (including yeasts). These have been used as thickeners, gelling agents, encapsulation and film-making agents or stabilizers (Giavasis, 2013, pp. 413–468). Although some of the most studied polysaccharides produced by mushrooms (*e.g.* schizophyllan and lentinan) are already available and marketed as nutraceuticals (pharmaceutical formulation), their addition to food in their purified form has not been commercialized as yet (Giavasis, 2013, pp. 413–468). Still, in the context of bakery products some studies document the incorporation of mushroom polysaccharides in foods containing cereals (Foschia, Peressini, Sensidoni, & Brennan, 2013; Ktenioudaki & Gallagher, 2012). In an attempt to produce a novel high-fibre and low-calorie functional food, Kim et al. (2011) swapped part of the wheat flour in baked foods for glucans from *L. edodes*. These glucans (included in batter at a concentration of 1 g of β-glucan per serving) improved the pasting properties of wheat flour and increased batter viscosity and shear-thinning elasticity without any adverse effect on air holding capacity or hardness.

L. edodes paste added to wheat flour for making noodles was also studied. This "recipe" resulted in a higher quality, fibre-rich functional food with antioxidant and hypocholesterolemic properties (Kim, Kang, & Kim, 2008; Kim, Chung, Nam, & Kang, 2009). *L. edodes* powder also proved to be effective as an oil barrier and texture-enhancing ingredient in frying batters (Kim et al., 2010).

B-glucans have also been tested for encapsulation of pickling liquid to be released in soups or sauces during cooking (Laroche & Michaud, 2007).

To develop a novel snack food, *Pleurotus sajor-caju* (Fr.) powder was incorporated in the preparation of "papad", a food adjunct or typical Indian snack. The papad enriched with 20% mushroom powder presented increased values of protein, minerals and crude fibre, giving rise to a new product with improved nutritional value and fibre contents (Parab, Dhalagade, Sahoo, & Ranveer, 2012). *A. bisporus* extracts were used to produce snacks with high antioxidant potential and free radical scavenging capacity. This work aimed to show that mushroom snacks may serve as a good alternative for currently existing snack foods since they retain a significant amount of polyphenolics and antioxidants, which constitute beneficial health effects to consumers. Moreover, the production process is quite simple and economically viable (Singla, Ghosh, & Ganguli, 2009).

Brennan, Derbyshire, Tiwari, & Brennan, 2012 and 2013, incorporated the freeze-dried powder from *Agrocybe aegerita* (V. Brig.) Singer as well as mushroom material prepared from stems discarded during mushroom production in cereal snack product, to obtain a healthy extruded snack. This increased product expansion and density while decreasing the potential glycemic response of food.

Other food types were also studied for mushroom incorporation namely cheese and cheese-related products. Okamura-Matsui et al. (2001) produced a functional cheese-like food incorporated with *S. commune*, which was shown to have preventive effects against cancer and thrombosis. Ribeiro et al., 2015 microencapsulated, the hydroalcoholic extracts from *Suillus luteus* (L.) Roussel and *Coprinopsis atramentaria* (Bull.) Redhead, Vilgalys & Moncalvo in cottage cheese by spray-drying. This incorporation resulted in synergistic effects between the extracts with respect to antioxidant activity. Moreover, the antioxidant properties were preserved over time and the final product maintained its nutritional properties and no colour modifications were observed.

In addition to the antitumour and antioxidant properties, improving taste and taking advantage of the nutritional value of incorporated products, the antimicrobial capacity has also been explored in some studies. Here, research evaluated the possibility of using natural extracts as food preservatives due to their antimicrobial potential, avoiding food deterioration. Barros et al. (2011) incorporated *Boletus edulis* Bull. extracts into beef burgers and concluded that this protected meat from lipid peroxidation. Another study using pork meat as a matrix provided a basis for developing new naturally-occurring antimicrobial agents to extend shelf life of meat products (Stojković et al., 2015). In this work, *Boletus aereus* Bull. extract proved that it could be used as an effective controlling agent against food-poisoning bacteria (*i.e. Staphylococcus aureus*, *Listeria monocytogenes*, *Escherichia coli* and *Salmonella typhimurium*). Reis et al. (2012) reported the inhibition of food contaminants namely *Penicillium verrucosum* var. *cyclopium* in cream cheese after incorporation of this product with *Agaricus bohusii* Bon extract. Another study with cream cheese, reported the beneficial effects after incorporation with *A. aegerita* (Petrović et al., 2015). Moreover and as in the previous study (incorporated with *A. bohusii*), this new product was well accepted by a group of taste panelists. *Laetiporus sulphureus* (Bull.:Fr.) Murr. also showed excellent preserving potential against the fungus *Aspergillus flavus* in preliminary studies with chicken pâté (Petrović et al., 2014). Also, the well-known champignon (*A. bisporus*) as well as the species

A. blazei, revealed good preserving properties in yoghurt against the pathogen *Listeria monocytogenes* (Stojković et al., 2014). The desert truffle, *Tirmania pinoyi* (Maire) Malençon from Libya, after incorporation in soup proved to be effective against *Staphylococcus aureus* (Stojković et al., 2013).

Although these results seems very promising, it should be kept in mind that, besides the consumption of foods rich in bioactive compounds appears to be beneficial, to claim that a functionalized food will bring beneficial effects on the consumers' health, needs more in-depth studies. For example, bioaccessibility and bioavailability studies are required, because of the changes occurred in the digestive tract. It is known that some bioactive compounds, for instance high molecular weight food polyphenols, are not bioaccessible in the human body. They pass unaltered through the upper intestine in association with the food matrix (Quirós-Sauceda et al., 2014).On the other hand, it has been reported that low molecular weight phenolics are released from the food matrix in the upper area of the gastrointestinal tract by direct solubilization in the intestinal fluids and/or by the action of digestive enzymes (Pérez-Jiménez et al., 2009), being at least partially absorbed through the small intestine mucosa. However, despite the bioavailability of the different phenolic compounds may be variable, it is generally established that after intake, this is relatively low (Heleno et al., 2015; Quirós-Sauceda et al., 2014). Moreover, insoluble and soluble dietary fibre have been suggested as interfering with the bioavailability/absorption of macronutrients and biomolecules, especially fat, and some minerals and trace elements in human digestion (Quirós-Sauceda et al., 2014). Therefore, being mushrooms especially rich in fibre, this may interfere with nutrients or bioactive compounds absorption. Consequently, some studies are needed in order to obtain such answers. However, even if absorption of bioactive compounds, *i.e.* antioxidant compounds, is sometimes reduced, its consumption is still an advantage because it promotes a healthy antioxidant environment (Quirós-Sauceda et al., 2014).

In fact, some studies have been made regarding the capability of some metabolites obtained after ingestion of phenolic compounds to reduce, maintain or increase the bioactivity of the parental molecules. *In vivo*, these compounds are metabolized and circulate in the organism as glucuronated, sulphated and methylated metabolites (Heleno, Martins, Queiroz, & Ferreira, 2015b). It has been described that while ferulic and caffeic acids, the two phenolic acids most representative in the human diet, are extensively metabolized after absorption. However, their glucuronated metabolites can retain a strong antioxidant activity and might still exert a significant antioxidant action *in vivo* (Heleno et al., 2015b). Studies on the antimicrobial activity of some mushroom phenolic compounds after undergoing the methylation reactions have also been reported. It has been concluded that reactions have considerably increased the activity of *p*-coumaric acid. Moreover, the inclusion of acetyl groups increased the antifungal activity while the antibacterial effects were maintained. The maintenance of the antitumour potential of phenolic compounds after ingestion has been also reported, and in most of the cases the substitution of the carboxylic group (in parental organic acids) for an ester (in methylated derivatives) increased the cytotoxicity activity of the compounds (Heleno, Martins, et al., 2015). Glucuronated derivatives also showed an increased cytotoxicity comparing with the corresponding parental compounds, due to the inclusion of acetyl molecules (Heleno, Martins, et al., 2015).

Despite the studies already developed in this field of mushroom incorporation in foods, these products have not yet gone to market. As mentioned before, although some works with phenolic compounds have been carried out, there is still a lack of information regarding the bioaccessibility/bioavailability of the compounds and possible interactions with the food matrix. Therefore, most of the mushrooms and their compounds are mainly consumed in their natural form or in dietary supplements.

3. Mushrooms as a source for nutraceutical development

3.1. Bioactive compounds in mushrooms used as nutraceuticals

Due to the bioactive potential of mushrooms and the growing interest in traditional medicines for "treating" various physiological disorders, a new and specific term has now been recognized: "mushroom nutraceuticals" which should not be confused with the general nutraceuticals, functional foods and/or pharmaceuticals. A mushroom nutraceutical is considered a refined or partially refined extract or dried biomass from either mycelium or the fruiting body of a mushroom which is consumed in the form of pharmaceutical formulations, capsules or tablets, as a dietary supplement and has potential therapeutic applications (Giavasis, 2014;; Wasser, 2014).

The principal nutraceuticals found in mushrooms include: i) lipids, especially unsaturated fatty acids; ii) vitamins, such as vitamin E and vitamin C; iii) proteins, peptides and amino acids, including lectins, leucine and valine; iv) carbohydrates, especially polysaccharides, such as lentinan. Due to the existence of a wide variety of bioactive compounds in this matrix, this has great potential to produce new nutraceutical formulations. Although recognizing the great potential of mushrooms as the basis of such formulations, some unresolved problems remain which affect their preparation and subsequent marketing such as safety issues, standardization, regulation, efficacy and mechanism of action (Wasser, 2014). Therefore, and despite interest in such products having grown over the years, the market in Asian countries has been exploited more. In the West, this remains a market requiring greater investment and exploitation (Giavasis, 2014).

Although somewhat contradicting that previously stated, one of the advantages of using mushroom-based dietary supplements (DSs) is related to their safety. The majority of mushrooms used in DSs are cultivated commercially (and not gathered in the wild). Therefore, no doubts exist regarding their taxonomic identification and they exclude adulterated products. Besides, since mushrooms are easily propagated *in vitro* it is possible to maintain a single clone for all production. Another great advantage (especially for those species that cannot produce fruiting bodies artificially) is that they are capable of growing in the form of mycelial biomass in submerged cultures and are utilisable (Wasser, Sokolov, Reshetnikov, & Timor-Tismenetsky, 2000).

Although there are some barriers to the marketing of these products, it is possible to find some formulations on the market.

3.2. Formulations available on the market and success rate

Today, some dietary supplements based on mushrooms are available on the market. These include i) artificially cultivated fruiting body powders, hot water or alcohol extracts of these fruiting bodies; ii) dried and pulverized preparations of the combined substrate, mycelium and primordial mushroom; iii) biomass or extracts from mycelium harvested from submerged liquid culture grown in a fermentation tank or bioreactor; iv) naturally grown, dried mushroom fruiting bodies in the form of capsules or tablets; and v) spores and their extracts (Wasser, 2014).

Table 2 lists some worldwide nutraceutical formulations available on the market and purpose of their use, including some scientific research made with some of the commercially available formulations.

From the available formulations, we highlight some of the most widely known nutraceutical formulations based on some of the

Table 2
Main mushroom-based products available on the market.

Product	Use (based on the manufacturer claims)	Mushroom species present in the formulation	Reference
Active hexose correlated compound (AHCC)	- Immunomodulatory agent, also effective in therapy for cancer and various infections	- *Lentinula edodes*	Drugs.com. AHCC Uses, Benefits & Dosage Herbal Database. https://www.drugs.com/npp/ahcc.html/Accessed 19.03.17
Agaricus blazei Capsules (or Royal Sun Agaricus or ABM)	- Immune support supplement	- *Agaricus blazei* - *Cordyceps* spp.	Aloha Medicinals. *Agaricus blazei*. http://www.alohamedicinals.com/agaricus-blazei.html#.V2iDO9IrLI Accessed 19.03.17
Agarikon.1	- Enhancement of immune system	- *Agaricus blazei* - *Ganoderma lucidum* - *Grifola frondosa* - *Lentinula edodes* - *Pleurotus ostreatus*	Myko San Health from Mushrooms. Agarikon.1, Professional medicinal mushrooms extract for cancer patient support. http://mykosan.com/agarikon-1-medicinal-mushrooms-cancer/ Accessed 19.03.17
BreastDefend®	- Immune support - Hormonal support	- *Ganoderma lucidum* - *Phellinus linteus* - *Trametes versicolor*	ecoNugenics. BreastDefend. https://www.econugenics.com/breastdefend/ Accessed 19.03.17
Breast-Mate®	- Support healthy breast tissue	- *Grifola frondosa* - *Phellinus linteus*	Mushroom Wisdom. Breast-mate. http://www.mushroomwisdom.com/products_detail.php?product_id=38&productCat=breast-mate&maitake_id=27tjonr2g3bn3ncmqep8e0b5v2 Accessed 19.03.2017
Concord Sunchih GPSP	- Supports and enhances the immune system; - Helps in the improvement of general wellbeing; - Relief from allergy or hay fever symptoms; - Temporary cough relief; - Relief from pre-menstrual symptoms/syndrome; - Relief from sleeplessness; - Temporary relief from rheumatoid arthritis pain; - Relief from influenza/flu symptoms; - Relief from nervous tension, stress and mild anxiety; - Temporary migraine headache relief; - Maintaining health of capillaries; - Relief from menopausal symptoms; - Aids or assists in the maintenance of peripheral circulation; - Helps support a healthy heart, normal healthy blood pressure, normal healthy cholesterol levels, normal blood sugar levels, and a healthy respiratory system; - Antioxidant	- *Ganoderma lucidum*	Concord Mushroom Supplements. Sunchih GPSP. http://www.concordhealth.net/product/-sunchih-gpsp Accessed 19.03.2017
Coriolus MRL™	- Immune system support	- *Trametes versicolor*	Mycology Research Laboratories. Coriolus-MRL. http://www.mycologyresearch.com/products/coriolus Accessed 19.03.2017
Didanosine (Videx and Videx EC)	- Used along with other medications to treat human immunodeficiency virus (HIV) infection, through the decrease of HIV in the blood	Synthetic compound (based on the *Cordyceps* spp. compound)	University of California, San Francisco. Susa Coffey, MD, Didanosine (Videx). http://hivinsite.ucsf.edu/InSite?page=ar-01-02 Accessed 19.03.2017
GanoPoly®	- Helpful for assisting recovery after illness; - Supports healthy nervous system and quality of sleep; - Promotes calmness and sense of well-being; - Beneficial for healthy menstrual function; - Supports health of the heart and blood vessels; - Supports a healthy immune system; - Protects the body against cellular damage; - Helps support normal cholesterol balance and blood sugar levels; - Supports liver health.	- *Ganoderma lucidum* only or combined with *Cordyceps sinensis*	Gano Shopping. GanoPoly. http://www.ganoshopping.com/category.php?id_category=5 Accessed 19.03.2017
Imunoglukan P4H® capsules	- Immune system enhancement	- *Pleurotus ostreatus*	Imunoglukan P4H. Imunoglukan. http://www.imunoglukan.com/en/products Accessed 19.03.2017

Table 2 (*continued*)

Product	Use (based on the manufacturer claims)	Mushroom species present in the formulation	Reference
LifeShield™ Immune Support	- Immune system support	- *Ganoderma lucidum*	New Chapter. LifeShield Immune Support. http://www.newchapter.com/mushrooms/lifeshield-immune-support Accessed 19.03.2017
Mycoformulas Endurance™	- Enhancement of intracellular energy exchange, increased oxygenation and natural endurance	- *Cordyceps sinensis*	Myco Formulas. Mycoformulas Endurance. http://mycoformulas.com/endurance-mushroom-supplements.html Accessed 19.03.2017
Mycoformulas Immune™	- Improvement in immune system	- *Ganoderma lucidum*	Myco Formulas. Mycoformulas Immune. http://mycoformulas.com/immunity-mushroom-supplements.html Accessed 19.03.2017
MycoFormulas Memory™	- Memory support and daily mental clarity	- *Hericium erinaceus*	Myco Formulas. Mycoformulas Memory. http://mycoformulas.com/memory-mushroom-supplements.html Accessed 19.03.2017
MycoPhyto® Complex	- Support immune function	- *Agaricus* spp. - *Cordyceps* spp. - *Coriolus* spp. - *Ganoderma lucidum* - *Grifola frondosa* - *Umbellatus* spp.	ecoNugenics. MycoPhyto Complex. https://www.econugenics.com/mycophyto-complex-main/ Accessed 19.03.2017
MycoPlex-7™	- Promote health and vitality	- *Agaricus blazei* - *Cordyceps sinensis* - *Hericium erinaceus* - *Poria cocos* - *Schizophyllum commune* - *Trametes versicolor* - *Tremella fuciformes*	Gourmet Mushrooms. Mushroom nutraceuticals: A positive approach to health & wellness. MycoPlex-7™. http://gmushrooms.com/health/ Accessed 19.03.2017
Mykoprotect.1	- Immune system improvement	- *Ganoderma lucidum* - *Lentinula edodes*	Myko San Health from Mushrooms. Mykoprotect.1, Professional medicinal mushroom extract for viral infections. http://mykosan.com/mykoprotect-1-antiviral-medicinal-mushroom-extract/ Accessed 19.03.2017
ORIVEDA® PSP-50	- Immune supporter	- *Trametes versicolor*	Oriveda. ORIVeDA PSP-50 extract. http://www.oriveda.com/coriolus_psp.php Accessed 19.03.2017
Organic Rare Red Reishi™	- Promote good health and longevity through the positive effects on the immune system (including antitumour potential)	- *Ganoderma lucidum*	Gourmet Mushrooms. Mushroom nutraceuticals: A positive approach to health & wellness. Organic Rare Red Reishi™. http://gmushrooms.com/health/ Accessed 19.03.2017
Organic ReiShi-Gen	- Promote good health and vitality through the antitumour, antibiotic and antiviral properties, as well as the presence of antioxidant, immune-system stimulants, blood pressure regulators and anti-cholesterol agents	- *Ganoderma lucidum* - *Lentinula edodes*	Gourmet Mushrooms. Mushroom nutraceuticals: A positive approach to health & wellness. Organic ReiShi-Gen http://gmushrooms.com/health/ Accessed 19.03.2017
Organic Trimyco-Gen™	- Promote health and vitality	- *Cordyceps sinensis* - *Ganoderma lucidum* - *Lentinula edodes*	Gourmet Mushrooms. Mushroom nutraceuticals: A positive approach to health & wellness. Organic Trimyco-Gen™. http://gmushrooms.com/health/ Accessed 19.03.2017
OsteoMykon	- Maintenance of bone health (indicated for osteoporosis)	- *Grifola frondosa* - *Lentinula edodes*	Myko San Health from Mushrooms. OsteoMykon, Bone Health and Osteoporosis. http://mykosan.com/osteomykon-osteoporosis-bone-health/ Accessed 19.03.2017
ReishiMax GLP®	- Immune system support; - Protects liver function; - Maintains normal blood glucose levels; - Improves the quality of sleep and enhances overall well-being	- *Ganoderma lucidum*	Nu Skin. ReishiMax GLP® https://www.nuskin.com/en_ID/products/pharmanex/shop_all_categories/immune_support/41003519.html Accessed 19.03.2017

(*continued on next page*)

Table 2 (*continued*)

Product	Use (based on the manufacturer claims)	Mushroom species present in the formulation	Reference
Grifon Pro D-Fraction	- Enhancing the cellular immune system	- *Grifola frondosa*	Healing Edge Sciences, Inc. Grifon Pro D-Fraction. http://www.healingedge.net/store/product1360.html Accessed 19.03.2017
MaitakeGold 404®	- Providing daily immune protection	- *Grifola frondosa*	MaitakeGold 404®. http://maitakegold.com/ Accessed 19.03.2017
Lentinan/ LentinanXP/Lentinex®	- Used as an adjuvant of chemotherapy (especially in patients with gastric and colorectal carcinoma); - Upregulation of the immune system	- *Lentinula edodes*	Drugs.com. Lentinan. https://www.drugs.com/npp/lentinan.html Glyca Nova. LentinanXP/Lentinex® http://glycanova.com/products/nutraceuticals/lentinanxp-lentinex/ Accessed 19.03.17
PSK-16™	- Immunomodulating Support	- *Agaricus blazei* - *Armilleria mellea* - *Cordyceps militaris* - *Cordyceps sinensis* - *Ganoderma applanatum* - *Ganoderma lucidum* - *Grifola frondosa* - *Grifola umbellata* - *Hericium erinaceus* - *Lentinula edodes* - *Poria* spp. - *Poria sclerotium* - *Trametes versicolor* - *Tremella* spp. - *Pleurotus ostreatus* - *Schizophyllum commune*	Biogenuine. PSK-16™. http://www.biogenuine.com/ Accessed 19.03.17

Product	Use/Bioactivity (based on scientific claims)	Mushroom species present in the formulation	Reference
Agaricus blazei extract (ABM, obtained from the mycelium of the mushroom)	Hypoglycemic/Anti-diabetic	- *Agaricus blazei*	Hsu, Liao, Lin, Hwang, & Chou, 2007
Active hexose correlated compound (AHCC – produced from the mycelia of *Lentinula edodes*, marketed in the U.S. as an alpha-glucan rich dietary supplement)	Antiviral	- *Lentinula edodes*	Nogusa, Gerbino, & Ritz, 2009 Ritz, Nogusa, Ackerman, & Gardner, 2006
Ganopoly® (polisaccharidic extract from *G. lucidum*)	Antiviral	- *Ganoderma lucidum*	Zhou, Gao, & Chan, 2005
Hispidin (polyphenol naturally occurring in *Phellinus* species)	Anti-inflammatory	- *Phellinus spp.*	Shao, Jeong, Kim, & Lee, 2015
Hispolon (polyphenol naturally occurring in *Phellinus linteus*)	Anti-inflammatory	- *Phellinus linteus*	Chang et al., 2011
Immune Assist™ (immune-active heteropolysaccharides and β-glucans concentrated and purified from six mushroom species: *C. sinensis*, *A. blazei*, *L. edodes*, *G. frondosa*, *G. lucidum*, and *T. versicolor*. It is sold in the form of immune supplement.)	Antiviral	- *Agaricus blazei* - *Cordyceps sinensis* - *Ganoderma lucidum* - *Grifola frondosa* - *Lentinula edodes* - Trametes versicolor	Adotey, Quarcoo, Holliday, Fofie & Saaka, 2011
SX-Fraction® mushroom supplement (dietary supplement obtained from *Grifola frondosa*)	Hypoglycemic/Anti-diabetic	- *Grifola frondosa*	Konno, 2001 Konno, Alexander, Zade, & Choudhury, 2013

best known medicinal mushrooms. *G. lucidum* is one of the most widely known medicinal mushrooms which have been consumed for many years in traditional East Asian medicine as a dry powder or hot water extract (a type of bitter mushroom infusion). This has been recognized as a popular supplement taken by healthy individuals to boost their immune system and by cancer patients along with conventional therapies. GanoPoly®, a *G. lucidum* polysaccharide extract, is one of the best known DSs on the market.

β-glucans extracted/isolated from the fruiting bodies of the mushroom *G. frondosa* and commercially available as D-fraction, MD fraction and MaitakeGold 404® nutraceutical extract are available on the market (Point Institute, 2013). These are consumed essentially for immune protection.

Active Hexose Correlated Compound (AHCC), a α-glucan rich

nutritional supplement produced from the mycelia of shiitake (*L. edodes*) developed in Japan in 1989, is commercially available today. Legally, it is governed by the rules of functional foods (Matsui et al., 2002) and is known for its antioxidant (Ye, Ichimura, Wakame, & Ohe, 2003), anti-inflammatory (Ocón et al., 2013), immunomodulatory (Daddaoua et al., 2013), and antitumour (Matsui et al., 2002;; Shah et al., 2011) effects.

As previously referred, lentinan, a β-glucan isolated from shiitake mushroom is also widely known for its immunomodulatory/anticancer properties and is administered intravenously as an anticancer agent in Japan. In occidental countries, it is possible to find a superfine dispersed powder of lentinan which is used for oral administration (Lentinex® in Europe and LentinanXP in USA), also in an attempt to take advantage of its immune-enhancing potential (Point Institute, 2013). Other uses have been made with respect to this sample, including its use as a food ingredient. The safety of Lentinex® as a novel food ingredient has been established at specific proposed conditions of use and the proposed levels of intake (intake of 2.5 mL Lentinex® containing 1 mg lentinan (β-glucan)/mL correspond to 41.7 μg/kg body weight for a 60 kg person) (EFSA, 2010).

As abovementioned, PSK and PSP, both produced essentially from the cultured mycelia from *T. versicolor*, have been used for many years in Japan and China. In Japan, PSK is prescribed for use at a dosage of 3 g per day as an adjunct immunotherapeutic agent together with chemotherapy and/or radiotherapy and surgical treatments of colorectal and gastric cancers. In other parts of Asia, it is sold as an OTC product in pharmacies to be taken in 500 mg capsules, as an extract or as a tea (The PSK Information Foundation, 2006).

Other formulations exist on the market such as a new product for dementia, particularly Alzheimer's disease, based on a proprietary standardized extract that contains hericenones and amyloban (both from the *H. erinaceus* (Wasser, 2014).

Although no specific directive exists for the marketing of authorized products some guidelines should be followed, particularly with regard to consumer protection against potential health risks and to ensure that they are not provided with misleading information (EFSA, 2016; Commission Regulation (EU) No 119/2014). It is known that most of the major medicinal mushroom polysaccharides have been subjected to extensive preclinical toxicological studies (Smith, Rowan, & Sullivan, 2002). For example, PSP was subjected to a wide range of preclinical toxicological tests, proving no effect on ovarian follicular development, steroidogenesis, quality of oocytes, pregnancy or embryo development, and showed no evidence of genotoxicity. Also lentinan or other medicinal mushroom derivatives have not reported any serious adverse reactions or evidence of drug-drug interactions (Smith et al., 2002).

Consumer safety is a premise but for most compounds some issues are poorly understood, such as the interaction with other drugs. This aspect is quite important, particularly with respect to compounds used as co-adjuvants of antitumour therapies. Thus, this should be an area for future studies to be undertaken.

Based on the points mentioned above (*i.e.* with the products (DSs) available on the market being quite diverse but with few standard protocols to ensure their quality) this should be one of the priorities in this field. This would also contribute to increase consumer confidence (Sullivan, Smith, & Rowan, 2006).

4. Limitations and future perspectives

Throughout this review, we have seen the enormous potential of mushroom as health-promoting agents. However, there are still some limitations regarding their use/consumption, particularly the consumption of new foods incorporating mushrooms and the use of new bioactive formulations.

As abovementioned, one advantage of using formulations or compounds obtained from edible mushrooms is, in fact, their edibility. Therefore, they are quite safe (to some extent) regarding toxicity effects. Furthermore, some compounds already used, such as lentinan and schizophyllan have already been tested and revealed low or zero toxicity, even at high doses. Also, for other species like *A. blazei*, several safety studies have been carried out to establish their safe use as the basis for functional foods. This species is now considered a safe immunostimulant mediator of biochemical parameters related to obesity or diabetes (Yamanaka, Liu, Motoi, & Ohno, 2013).

However, there are some reports of diarrhea or bloating associated with shiitake consumption and some allergic reactions affecting the skin, nose, throat or lungs associated with other species. Nevertheless, no known side effects exist from eating normal amounts of mushrooms in our diet and regarding most of the available mushroom supplements, they have not been tested to establish if they interact with medicines, foods or other herbs and supplements (Cancer Research UK, 2013). Likewise, some studies, namely the explanation of the modes of action of each extract/compound should be taken into account, in order to verify possible interactions with other components or with other drugs. Moreover, the implementation of standard protocols should be established which ensure the quality of the products as well as greater acceptance and trust among consumers. These issues should be considered and in-depth studies undertaken in the near future, which would allow the market for functional foods and dietary supplements to grow in a rational and cohesive manner.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.tifs.2017.05.010.

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